

PhysicsElectricity

1.

Electric current:-

It is the flow of electrons through a conductor.

The device which causes the slow of electrons through a conductor is called a cell.

Electrons flow from the negative terminal to the positive terminal.

Electric current flows from positive terminal to the negative terminal.

The steady flow of charges through a conductor is the quantified charges flowing through a conductor in unit time.

$$I = \frac{Q}{t}$$

↳ current $t = \text{time}$

not change

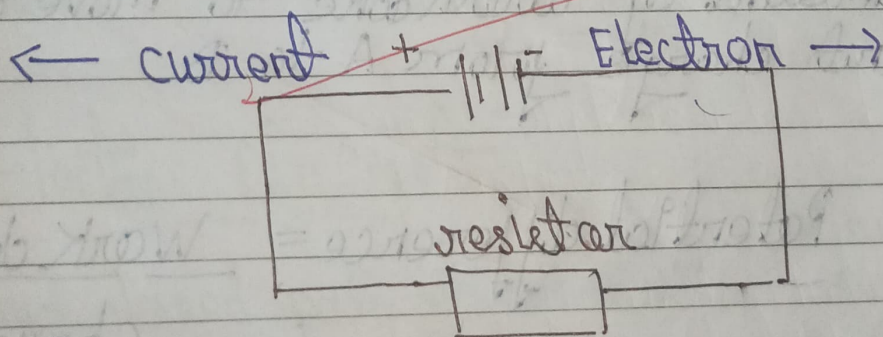
SI unit of charge is Coulomb (C).
 It is the charge contained in 6×10^{18} electrons.

SI unit of current is called Ampere (A)

One ampere is the current flowing through a conductor if 1 coulomb of charge flows through 1 second

$$1 \text{ ampere} = \frac{1 \text{ coulomb}}{1 \text{ second}}$$

Electric current is measured by Ammeter. It is always connected in series in a circuit



Potential difference :-

Electric current will flow if any conductor is provided or if there is a difference in electric potential between the two ends of the conductor.

This difference in electric potential between two ends of a conductor is called potential difference.

The potential difference in a circuit is provided by a cell or a battery. The chemical reaction in the cell produces a potential difference between the two terminals and sets the electrons in motion and produces electric current.

Potential difference: between two points A and B of a conductor is the amount of work done to move a unit charge from A to B.

$$\text{Potential difference} = \frac{\text{Work done}}{\text{charge}}$$

$$\text{OR } V = \frac{W}{Q}$$

SI unit of potential difference is measure. $\rightarrow$ Volt (V) $\rightarrow$

One volt is the potential difference when 1 Joule of work is

done to move a charge of 1 coulomb from one point to another.

$$1 \text{ volt} = \frac{1 \text{ Joule}}{1 \text{ coulomb}} \quad \text{or}$$

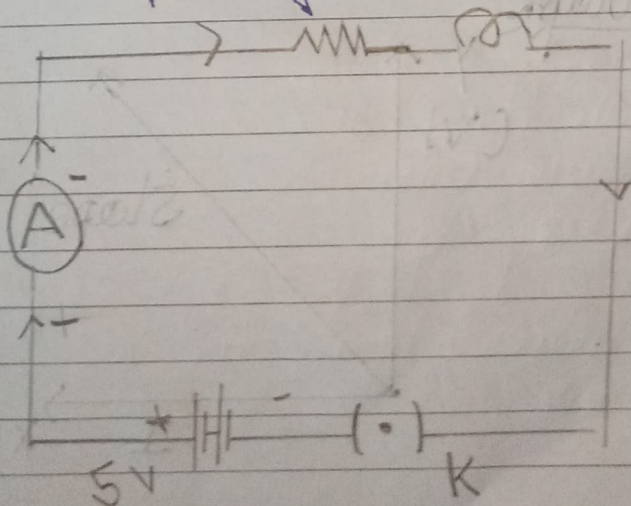
$$V = \frac{W}{Q}$$

Potential difference is measured by a voltmeter.

It is always connected in parallel across the two points between which the potential difference is to be measured.

Electric Circuit:-

is a continuous and closed path of an electric current.



Ohm's Law is a relationship between the potential difference across conductor and current flowing through it.

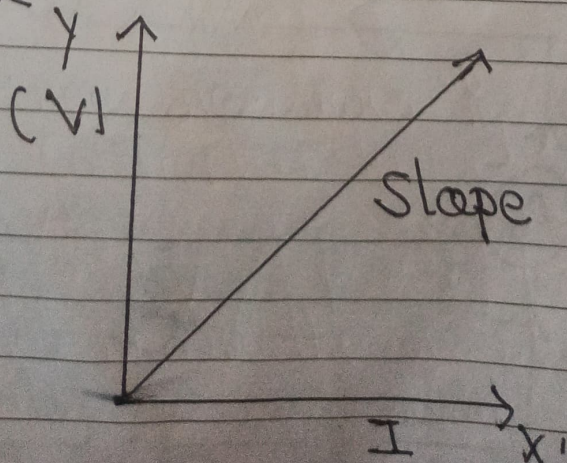
Ohm's law states that the current flowing through a conductor is directly proportional to the potential difference between its ends provided its temperature remains constant.

$$I \propto V \quad \text{or} \quad \frac{V}{I} = \text{constant} \quad \text{or} \quad V \propto I$$

$$\frac{V}{I} = R$$

$$[V = IR]$$

VI Graph



Date / /

$$\text{Slope} = \frac{y}{x} = \frac{V}{I} = R$$

[Slope = Resistance]

More the angle of the slope more the resistance

Resistance

Property of a conductor to resist the flow of current through it

According to Ohm's law $\Rightarrow R = \frac{V}{I}$

SI unit of resistance is ohm (Ω)

The current flowing through a resistor is inversely proportional to the resistance

So if the resistance is doubled then the current gets halved

Factors on which the resistance depends

Length (l)

Area of cross section (A)

Material of the conductor

$$R \propto l$$

$$R \propto \frac{1}{A}$$

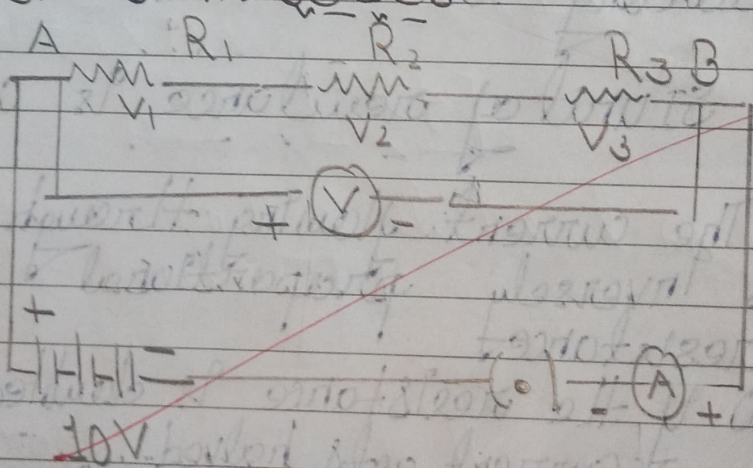
$$R = \rho \frac{l}{A}$$

where ρ (rho) is a constant of proportionality called resistivity of the materials of the conductor.

SI unit of resistivity is ohm (meter) (Ωm)

[Metals < Alloys < Insulators]

Resistors in Series



When three resistors R_1 , R_2 and R_3 are connected in series across AB.

The current in all the resistors are same

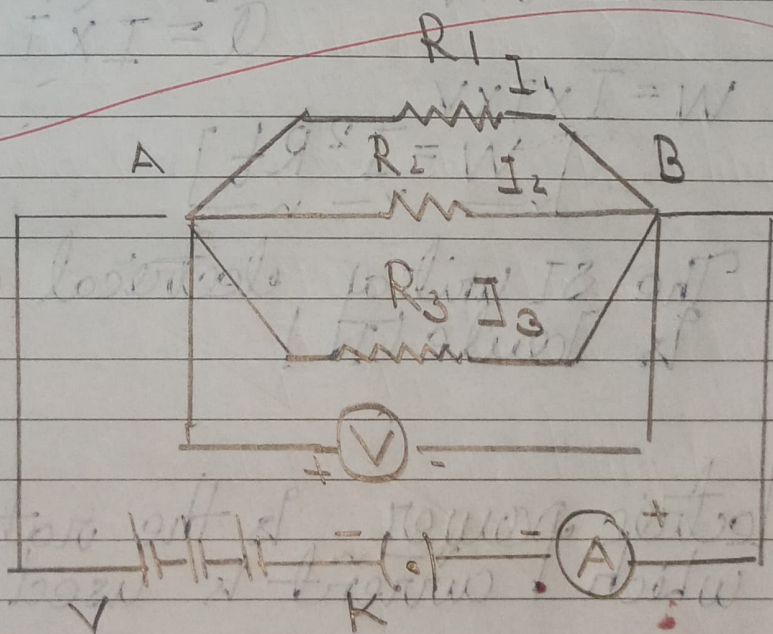
- The total voltage (PD) across the resistors is equal to the sum of the voltage across each resistor

$$V = V_1 + V_2 + V_3$$

- The equivalent resistance is the sum of resistance of each resistor.

$$R_e = R_1 + R_2 + R_3$$

Resistors in parallel



The PD in all resistors is the same

The total current in all the resistors is the sum of the current in each resistor.

$$I = I_1 + I_2 + I_3$$

The reciprocal of equivalent resistance is the sum of the reciprocals of each resistance.

$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Electrical energy is the work done to maintain the flow of current in a conductor.

$$W = Q \times V$$

$$I = Q / T$$

$$Q = I \times T$$

$$W = I \times T \times V$$

$$[W = I^2 R t]$$

The SI unit of electrical energy is Joule (J).

Electric power is the rate at which current is used.

$$\text{Power} = I^2 R \text{ or}$$

$$P = I \times V \quad [P = VI]$$

The SI unit is Watt (W).

One watt is the power when 1A of current flows through PD of 1V

$$1000 \text{ W} = 1 \text{ kW} \Rightarrow$$

$$1 \text{ kWh} = 1000 \text{ watt} \times 3600 \text{ seconds} \\ = 3.6 \times 10^6 \text{ Joules}$$

Joule's heating effect of electric current:

$$W = Q \times V$$

$$P = \frac{W}{t} = \frac{Q \times V}{t}$$

$$\frac{Q}{t} = I \quad \text{or} \quad P = V \times I$$

~~or heat energy supplied = $P \times t = V \times I \times t$~~

According to Ohm's law $\Rightarrow V = IR$

$$\text{Heat produced} = I^2 R t$$

Mafes
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